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When “conflict free” minerals go to war

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ABSTRACT

The article develops a simple but important argument: “conflict free” minerals are essential to the waging of contemporary war. This argument is substantiated over three main sections. First, we provide historical background to the idea of “conflict minerals” to show how they are narrowly associated with the violence of extraction and with consumer products (phones, electric vehicles, etc) in way that forecloses their use in weapons manufacturing and war further along the supply chain. Second, we draw from fieldwork in Rwanda and secondary sources to explicate the ways that minerals attain “conflict free” certification despite documented links with conflict in central Africa. Transparency in supply chains, we show, is carefully angled: issues of provenance (i.e., the movement of minerals to and in Rwanda) are obscured yet meticulous systems are in place to enable and trace the movement of minerals from Rwanda. In the third section, we focus on the supply of tin and tantalum from Rwanda to weapons suppliers and outline the use of those minerals in contemporary military hardware. In conclusion we sketch an agenda for future research on “conflict free” minerals that go to war.

In this article we develop a simple but important argument: “conflict free” minerals are essential to the waging of contemporary war. By “contemporary war” we refer specifically to a mode of warfare enabled by technologically advanced systems of surveillance, navigation, and weaponry that is generally practiced by the militaries of richer, interventionist states (e.g., US, UK, Israel, and allies). The violence of war in this “late modern” form is distributed via an assemblage of hardware and software – GPS satellites, pilotless aircrafts, high-tech jets and tanks, precision-guided munitions, remote and night sensing equipment – that have been developed as part of the wider societal digital revolution of recent decades (see Belcher, 2011; Gregory, 2011; Griffiths, 2022; Griffiths & Redwood, 2024; Jabri, 2016). Contemporary technologies in this period, it is widely understood, depend on the mining and commodification of certain “technology critical” minerals from the Earth’s crust whose material properties enable increased performance capacities in a range of products that connect the comforts of end-point users (e.g., consumers) to ecological and public health at sites of extraction (e.g., in Central Africa). It is by-now a familiar (though still unaddressed) critique that the smart phones, laptops, and electric vehicles that furnish the lives of some are only available because of violent labour conditions and (para)military conflict at a primary site of extraction, the eastern and southern provinces of the DRC (see e.g., Eichstaedt, 2011; Kara, 2023; Nest, 2011; Smith, 2021; Vogel, 2018, 2022). The idea of “conflict minerals” thus emerges as those that precipitate and fund warfare; and “conflict free” minerals is the remedial response with a putative objective of ensuring a more ethical supply chain in the production of consumer goods. Such “conflict free” minerals, as we will explicate however, are also essential to weapons manufacturers and deployed by advanced militaries as minerals that go to war.

From Iraq and Afghanistan to Gaza, Libya, and Lebanon, a story of 21st century warfare is one of primarily aerial bombardment executed by militaries that deploy an arsenal of “efficient”, “accurate”, and “smart” equipment. While more “traditional” materials – steel, aluminium, brass – remain integral to the production of such equipment, digital systems and increased aeronautic capabilities depend on a variety of other minerals: lithium batteries, cobalt cathodes, tantalum resistors, nickel capacitors, tin semiconductors, tungsten electrodes, and so forth (see Griffiths & Rubaii, 2025). A number of these minerals have come under scrutiny by scholars and lawmakers precisely because of their connections to war at the “upstream” end of the supply chain: tantalum, tin, tungsten, and gold (known collectively as “3 TGs”) – metals associated with war in the DRC – are now subject to legislation in China, the European Union, and United States that seeks to verify more ethical provenance. So far, however, the vast majority of public and scholarly debate has centred on consumer electronics such that the convoluted supply chains of Apple, Samsung, Tesla, and similar companies are placed under critical and legal spotlight (e.g., Deberdt & Le Billon, 2021; Diemer et al., 2022; Vogel & Raeymaekers, 2016), with scarce attention paid to the significant *downstream* involvement of the weapons industry and the subsequent end-point use of minerals in war. Our objective here is to reorientate geographies of conflict minerals so

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that weapons manufacturers, their suppliers and subsidiaries are also included in the ways conflict minerals are conceptualised. But this is more than a conceptual issue: the fact that weapons manufacturers have a hand in the violence of extraction should redouble our intellectual and practicable efforts to document and intervene in the disperse networks of exchange that make war possible.

To these ends, our approach here considers conflict minerals from a specifically "downstream" perspective that moves away from an emphasis on the provenance of minerals for consumer goods to the militaristic use of raw materials in conflict settings distant from – yet intimately connected to – those that mark the conditions of extraction. The methodological objective is to work towards more comprehensive critiques of war that map temporally and spatially disperse geographies of military materials and military violence. Where scholars of militarism have undertaken this task in terms of the long durée of war and multiple aftermaths (e.g., Nixon, 2011; Pugliese, 2020), a complementing angle addresses the "beforemaths" that make war (and aftermaths) possible (see Griffiths & Rubaii, 2025). In practical terms, we work backwards (or "upstream") from the immediate and enduring violence of military operations to the production of the means of violence by tracing supply chains via ethnographic inquiry at sites of raw material production and detailed analysis of archival records of mineral commodification. Specifically, the information presented in this article is drawn from two main research activities: fieldwork in Rwanda, a key site in the exchange and legitimisation of "conflict free" minerals from central and east Africa where ore is imported, extracted, processed, smelted, and exported; and analysis of Specialized Disclosure Reports, the documents required under US law for companies (weapons manufacturers included) that use conflict minerals (3 TGs) in manufacturing processes. The account of conflict minerals that emerges from these activities teaches us that the question of conflict, ethics, and supply chains should not begin and end at the origins of minerals but must be extended to include their eventual use as means of war.

The article proceeds in three sections. In the first we provide historical background to the idea of "conflict minerals" and initiatives that are designed (it is claimed) to ensure the products of mining are not connected to ongoing war in the DRC. We also synthesise existing critique to show how the violence of extraction is narrowly associated with consumer products (phones, laptops, electric vehicles). As a result, we argue, multiple other trajectories of violence further along the supply chain – in the electronic devices that are deployed to make war – are foreclosed in current critical approaches to conflict minerals. The second section draws from both fieldwork in Rwanda and secondary sources to explicate the ways that minerals attain "conflict free" certification despite documented links with conflict in central Africa whereby minerals traded through Rwanda are likely Congolese in origin. Transparency in supply chains, we show, is carefully angled: issues of provenance – i.e., the movement of minerals *to* and *in* Rwanda – are obscured, yet records are in place to enable and trace the movement of minerals *from* Rwanda. In the third section, we focus on the supply of tin and tantalum from Rwanda to weapons suppliers and outline the use of those minerals in contemporary military hardware. To conclude, we reiterate the main arguments of the article – that "conflict free" minerals are essential to the waging of contemporary war and that the question of ethics should extend the full length of the supply chain – and sketch an agenda for further geographical research on "conflict free" minerals that go to war.

1. Conflict minerals: background and current framing

In this section we situate the idea of "conflict minerals" historically and geographically, its connection to the DRC and neighbouring states, and its narrow association with advanced tech industries involved in the production of consumer goods (phones, laptops, and electric vehicles). We demonstrate how "conflict minerals" has become a restrictive term that forecloses multiple trajectories of violence further along the supply chain, namely in the electronic devices that are deployed in war.

As a necessary preface, it is important to delineate often-conflated terms in discussions of minerals that are used in contemporary technologies. "Technology-critical elements" (TCEs) or "critical raw materials" (CRMs) are terms that are codified in United States[1] and European Union[2] policy that refer to periodically reviewed lists of 40–60 raw materials whose supply chains are "high risk" and that are crucial to modern and "green" technologies. "Rare earth elements" (REEs) refers a subset of technology-critical elements: 17 lanthanoid soft heavy metals that are considered similarly essential to 'high-tech consumer products, such as cellular telephones, computer hard drives, electric and hybrid vehicles, and flat-screen monitors and televisions'.[3] "Conflict minerals" is formally recognised by the EU, OECD, and US as a term that refers to tantalum, tin, tungsten, and gold and often appears in acronymised form as "3 TGs". These minerals are associated with the continuation of war in eastern and southern DRC where, for decades, armed groups (e.g., the state, mining companies, militia) have fought for control of territory that includes planet's richest (in commodifiable terms) geological formation. Cobalt is not classed as a conflict mineral, though it is essential for advanced technologies, especially to the production of lithium-ion batteries, and there are calls for it to fall under the legal frameworks of 3 TGs (e.g., Amnesty International, 2016, p. 44). Other more "traditional" metals such as copper are abundant in the DRC and remain crucial to electronic circuitry and are thus also linked to the perpetuation of war. For our purposes in this section, we use the term "conflict minerals" as shorthand to refer to the raw materials that are associated with war in the DRC (i.e., 3 TGs and other technology critical minerals such as cobalt and copper). In sections two and three we focus mainly on tantalum and tin, though our arguments apply to all minerals involved in conflict.

Depending on sources and specific years, it is estimated that the DRC accounts for ~70 % of cobalt, ~50 % of tantalum, ~9 % of tin, ~4 % of tungsten, and ~10 % of copper production worldwide. The greatly increased demand for these minerals from the early 2000s occurred in (and exacerbated) a context of relative lawlessness and general poverty in the eastern and southern provinces of the DRC (Haut-Katanga, Ituri, North and South Kivu, Lualaba, Maniema and Tanganyika), producing an economy of mining characterised by unregulated and pronounced exploitation. In a now familiar story, the production of minerals that power the lives of billions of people depends on coercive and violent conditions for those – including, importantly, the differently vulnerable bodies of children and women – who extract them from the ground. Child labour, enslavement, sexual violence, murder – as well as and ecological and epidemiological ruin – are commonplace, and even central to the continued supply of minerals at prices that maintain profitability further along the supply chain (Banza Lubaba Nkulu et al., 2018; Leon-Kabamba et al., 2018; Van Brusselen et al., 2020). Such conditions have given rise to intense scholarly (e.g., Nest, 2011; Smith, 2021; Vogel, 2018, 2022) and journalistic (e.g., Eichstaedt, 2011; Kara, 2023; Pitron, 2022) attention that informs a growing awareness by consumer publics. From early stories of "tin slaves" (Garrett & Ford-Robertson, 2008) and the "illicit mineral trade" (Rice & Borger, 2009) to mainstreamed and transformative campaigns (e.g., those run by the Enough Project, Fairphone, Global Witness), the idea of "conflict minerals" takes full form, becomes a signifier to mobilise ethical consumerism, and eventually political change in the form of legislation

[1] US Department of United States Department of Energy (2025) 'Critical Minerals and Materials' https://www.energy.gov/eere/ammto/critical-minerals-and-materials.

[2] European Commission (2025) 'Critical Raw Materials' https://single-market-economy.ec.europa.eu/sectors/raw-materials/areas-specific-interest/critical-raw-materials_en.

[3] US Geological Survey quoted here: https://www.americangeosciences.org/critical-issues/faq/what-are-rare-earth-elements-and-why-are-they-important.

around supply chain traceability and “conflict free” certification.

With important precedents, particularly the role of “conflict diamonds” (popularised also as “blood diamonds”) in Angola and Sierra Leone in the late 1990s, and subsequent legal frameworks (prominently: UN Security Council Resolution 1173 and the Kimberley Process Certification Scheme)[4] (see Le Billon, 2008; Zulu & Wilson, 2009), scholarly analysis of conflict minerals is mainly concerned with documenting upstream conditions and applying pressure downstream in terms of more ethical procurement. Scholarship that assesses or evaluates conflict mineral supply chains, mostly written from business and management studies perspectives, shows how large companies downstream create narratives of supplier engagement to make claims on transparency (Young et al., 2019) and the ways that supply chain disclosure reports to the US Security and Exchange Commission (that we discuss further below) allow manufacturers to sidestep questions of provenance by demonstrating minimal commitment to due diligence (Kim & Davis, 2016). A main conclusion of policy-orientated research is that regulations around conflict minerals – established in response to transnational human rights pressure – fall short of breaking the relationships between the financing of war and the control of mining sites and trade routes in the DRC and neighbouring states (see also Hanai, 2021; Kapoor, Pratt-Rogers, & Kahraman, 2022; Schwartz, 2016). A recurring concern across commentaries is that campaigns and certifying schemes emphasise – and thus attribute agency to – Western manufacturers and consuming publics, leaving little decision-making room left for Congolese and other actors at the extractive origins of supply chains (see e.g., Nest, 2011, p. 159). From this perspective, in so far as “conflict free” certification protects (certain) flows of capital, withdrawn trade with non-certified sources can expose mining communities to heightened labour precarity with the effect of often increasing levels of military violence where mineral flows are pushed further into informal (“illicit”) channels (Eichstaedt, 2011, p. 202; Kinniburgh, 2014, p. 65). At the sharpest end of critique, this dynamic of protected capitalist interests is interpreted as a form of “white saviourism” that is grounded in Orientalist ‘narratives about far-flung conflict zones’ that ‘mobilise a mercantilist strategy … to forcefully incorporate, embed and streamline extractive reservoirs and expendable bodies’ (Vogel 2022, 8). The idea of “conflict free” minerals is thus critiqued as a fiction of capital, an initiative of corporate social responsibility with largely negative effects on living conditions in mining areas.

Juxtaposing these conditions with the comforts produced further along the supply chain is a key rhetorical move that many scholars and journalists have made. There is an important geography fleshed out here where technological innovation and tech-related profits articulate with de-development in mineral producing areas, as Andreas Diemer and colleagues (2022, p. 10) explicate: ‘a significant spatial disparity between the locations where large amounts of CCMs [Critical and Conflict Minerals] are extracted and those where most of CCM-based technological returns are appropriated’. Investigating precisely this disparity, Christoph Vogel and Timothy Raeymaekers (2016, 1115) emphasise that the last decades have brought the ‘violent capitalist expansion in the margins of formalised global mineral markets’ in the DRC, referencing the ways that conflict minerals campaigns and legislation have – despite claims of success – in fact legitimised even “illicit” (i.e., conflict-connected) modes of extraction for market exchange. The account of “conflict free” minerals from this perspective is of economic monopolisation and capitalist accumulation that depend on the severe precaritisation of ecologies and livelihoods in mining areas (see also Calvão & Archer, 2021; Deberdt & Le Billon, 2021). The story here is of entrenched colonial geographies where 21st Century commodities have replaced rubber, ivory, and hardwoods as the products of labour and nature exploitation in Central Africa. In James Smith’s (2022) ranging and deep ethnography of the mining industry in the DRC, he connects decades of colonial and postcolonial war, extreme poverty and exploitation, and extraction to Silicon Valley and the “posthuman age”. Importantly, Smith (re)figures conflict minerals as “digital minerals” to accent their importance to the technological innovators of northern California.

Across this variously normative, evaluative, and critical work is a clear focus on, in the first instance, “multinational electronics firms” taken as exclusively consumer goods companies (phone and electric vehicle manufacturers are the most frequently mentioned), and in the second, a strong emphasis on the need to push ethical scrutiny upstream, towards the origins of conflict minerals. It is here that we identify a narrowness in the ways “conflict minerals” are addressed in the majority of existing scholarly inquiry: “tech” is made synonymous with consumer products even in an era of significant technology critical mineral use in weapons manufacturing; and the question of ethics and human rights is posed only at the extractive origins of minerals, thereby overlooking the rest of the supply chain. Given what we know of contemporary weapons companies – that they too are dependent on the very same minerals – some of this existing scholarship can be re-angled to ask slightly different questions that might get at broader relations between minerals and conflict.

Vogel and Raeymaekers (2016, p. 1115), for example, make the astute observation that ‘the way “conflict-free” sourcing is rolled out into the frontier [i.e., the DRC] … mistakenly assumes a linear commodity chain “from mine to market” influenced by static conflict dynamics, i.e. once a mine is “clean” or “green”, it will remain so’. Though their main concern is on the mutability of certain parts of the upstream supply chain (mainly mines and smelters) the point stands for our concern here: even if minerals can be “conflict free” at origin, once on a market composed of consumer goods producers *and* weapons manufacturers, there is no safeguarding against future use in war. Given also what we know of recent and ongoing wars – e.g., in Afghanistan, Gaza, and Iraq – we can also think more expansively and critically about the frames of conflict minerals critique. For instance, in his important investigation into cobalt mining in the DRC, based on a catalogue of harrowing mining injuries where miners young and old are exposed to horrific labour conditions coupled with a largely indifferent consuming public, Siddarth Kara (2023, 240) concludes: ‘this was the *final truth* of cobalt mining in the Congo: the life of a child buried alive while digging for cobalt counted for nothing’ (emphasis added). Kara’s evocative words are no less important if we consider also conditions in Afghanistan, Gaza, Lebanon, Iraq, and Ukraine and the fact that the cobalt mined by children in collapsing mines is part of the explosive capacities of militaries that collapses buildings in Fallujah, Jabalia, Baalbek, and Rafah on the bodies of children there. We might thus reframe to add that Kara’s image is not a “final truth” but a first one, followed by too many similar scenes further along this bloody supply chain.

2. The production of “conflict free” minerals

In this section we document the production of “conflict minerals” in legal and material terms. We begin by discussing relevant legislation and the formal processes that ground “conflict free” claims with reference to the case of Rwanda, a pivotal site in the supply of technology critical minerals. Information on Rwanda is compiled from both secondary sources and fieldwork conducted in late 2024 where we (AUTHORS 1 and 3) spoke with industry consultants, spokespersons, and managers at

[4] UN Security Council Resolution 1173 prohibits ‘the direct or indirect import from Angola to their territory of all diamonds that are not controlled through the Certificate of Origin regime of GURN; (c) to prohibit the sale or supply to persons or entities in areas of Angola to which State administration has not been extended of equipment used in mining services’ (see https://digitallibrary.un.org/record/255406?ln=en&v=pdf#files). The Kimberley Process is ‘is a commitment to remove conflict diamonds from the global supply chain’ that is backed by the UN and covers—it is claimed—more than 98 % of all diamond production (https://www.kimberleyprocess.com/).

mines, smelters, company offices, and transportation centres linked to tin and tantalum mining infrastructures. Piecing together different information, we show how minerals – despite documented links with conflict in central Africa – attain "conflict free" certification that enables their movement along supply chains towards end-point manufacturers. We identify a system of carefully angled transparency where issues of provenance – i.e., the movement of minerals *to* and *in* Rwanda – are obscured yet meticulous records are in place to enable and trace the movement of minerals *from* Rwanda.

In legislative terms, the EU Conflict Minerals Regulation (2021) and the US Dodd-Frank Act (2010) respond to raised awareness of the conditions of extraction of technology critical minerals documented in the previous section. Section 1502 of the Dodd-Frank Act, importantly, requires manufacturers that use technology critical minerals to submit "Specialized Disclosure Reports" to the Securities and Exchange Commission (SEC) with a specific focus on conflict in the DRC. Tantalum, tin, tungsten, and gold – i.e., 3 TGs – are covered by the legislation and companies must declare suppliers so far as they are able to identify them, often to the level of processing and smelting plants (and thus not mining sites themselves). The DRC and bordering states (Angola, Burundi, Central African Republic, Republic of the Congo, Rwanda, South Sudan, Tanzania, Uganda, and Zambia) together make up a legally defined region referred to as Conflict-Affected and High-Risk Areas, or CAHRA. Sourcing "conflict free" 3 TGs from CAHRA states requires suppliers' compliance with certificating schemes such as the Responsible Mining Initiative[5] or the ITRI Tin Supply Chain Initiative[6] that operate at different points of supply chains to audit mines and processing plants according to the OECD (2016) Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas. In collaboration and/or competition with these large-scale schemes, an array of smaller industry- and NGO-led certification initiatives has also formed in the CAHRA region such that conflict free certification can be said to have become an industry in itself (for an overview see Nest, 2011, pp. 105–159; Smith, 2021, 267–297).

In the growing market for "conflict free" certified minerals, the Rwandan government and mining industry has sought to capitalise on a specific geopolitical position: expansive and porous borders with the DRC and eastwards towards ports in Dar es Salaam (Tanzania) and Mombasa (Kenya); a long-held ambition to be recognised as a significant supplier of minerals for international markets; and a general credibility as a "democratic" and Western-facing state that is held up as a model of post-war reconstruction and sustainable development.[7] Minerals are Rwanda's largest export revenue earner – generating $1.1bn in 2023 (increased from $373m in 2017) – and are identified by the Rwandan Development Board (RDB) as a key area of future economic growth and international trade partnership (Visit Rwanda, 2025). A 'policy transformation' over recent years has sought to develop an 'enabling environment for mining industry', including establishing the Rwanda Mining, Gas, and Petroleum Board (RMB) in 2017 with a remit to realise the government's objective of becoming a 'mineral value-addition hub in the region' (RMB, 2025). In light of international regulation around conflict and mining, RDB and RMB strategy forefronts traceability in the Rwanda's extraction and processing activities, a strict adherence to OECD due diligence guidance, and a confidence in the final exportable commodity: 'minerals from Rwanda are 100 % conflict-free and traceable from mines to international markets' (RDB 2025). The RMB makes the claim that Rwanda is 'one of the most advanced countries' in the chain of custody procedures, highlighting partnerships with two organisations that provide mineral traceability services, ITSCI (International Tin Supply Chain Initiative) and RCS Global (RDB 2025). Both organisations audit different points in supply chains with a focus on conflict financing, human rights abuses, and corruption.[8]

Without space to elaborate further on the many initiatives, guidelines, and acronyms of conflict minerals policy in Rwanda, what is important to note is that certification schemes work within OECD (2016, 3) due diligence guidelines whose overall 'objective is to help companies respect human rights and avoid contributing to conflict through their mineral sourcing practices'. In practice, mineral sourcing in Rwanda is opaque and mining industry actors in central and east Africa are generally – and notoriously (see e.g., Smith, 2021, 269; Vogel, 2022, p. 115) – closed to external scrutiny. This was evident during our fieldwork in Rwanda where we sought to better understand the processes by which minerals attain "conflict free" certification before export. A first and important observation is that mining in Rwanda is highly sensitive and securitised; often even tentative inquiry was met with scepticism or even distrust. But it is important to stress any distrust was angled in a particular way: on the upstream of the supply chain rather than the downstream. In Kigali, for example, one mining consultant who agreed to meet with us, was keen to communicate Rwanda's global standing in technology metals exchanges, relaying that Rwanda's growing smelting infrastructures now export "to any company, to anywhere in the world". There was markedly less openness, however, about mining areas in Rwanda. Speaking back to market investigators and activists, the consultant – largely unprompted – insisted that Rwanda *does* in fact possess the geological riches of its neighbour, the DRC: "people ask *where are they really coming from?* … We have the same rock – minerals don't have boundaries!" He drew a map and described the border between Rwanda and the DRC, how the mineral deposits are the same on both sides: "really it is all the same rock!".

At another meeting, this time at a smelting facility close to Kigali that processes cassiterite for the global tin market, discussion around upstream and downstream of the supply chain followed a similar pattern. On this occasion, we were welcomed and taken through to a meeting room where a PR manager gave a broad overview of operations at the smelter, including the *realia* of allowing us to hold some raw cassiterite and a tin ingot. Most notable on the issue of supply chains was a certain geography of disclosure marked by a discrepancy in the willingness to provide details of the respective provenances and destinations of minerals. That is, there was an openness (and even pride) in emphasising downstream movements of minerals – "we ship to lots of places, depending on who orders it – it goes on a truck to the port in Tanzania then all over … Malaysia, China, Europe …" – but an entirely different disposition towards the upstream. "I do not want to say anything I shouldn't" were the very cautious words that responded to a (rather gentle) question on the smelter's chain of custody system and traceability. We were later able to establish that a part of the certification of "conflict free" minerals through the site depends on it taking on a *de facto* point of origin position whereby shipments *from* (yet not *to*) the site are digitally tagged and tracked. In this arrangement, the earthly origins of minerals are struck from the record while the multiple destinations – manufacturers of technology wares – are routinely documented. Such an approach to "conflict free" certification is instructive in the economies of what can and cannot be disclosed: there is a continued reticence to divulge proximity to conflict "upstream"; there is no such concern in the downstream supply chain. Whether for means of producing consumer goods or military equipment, "conflict free" minerals are free to travel.

This approach to transparency is also evident at further sites – ones that are certified as producers of conflict free tin, tungsten, and tantalum – where we spoke with management and workers where possible, asking questions around environmental policies and supply chain routes. On

[5] https://www.responsiblemineralsinitiative.org/.

[6] https://www.itsci.org/.

[7] One recent example of this reputation from an EU-Rwanda MOU on minerals supply: 'Rwanda with its favourable investment climate and rule of law can become a hub for value addition in the mineral sector' (European Commission, 2024).

[8] https://www.itsci.org/about-itsci/https://www.rcsglobal.com/bettermining/.

visits to multiple sites – for example, a wolframite (tungsten) mine and a cassiterite (tin) mine (both within 2 h drive north of Kigali), a tantalum smelting facility to the south, and different company headquarters in Kigali – there was again a general pattern of openness around onwards travel of final product alongside a pronounced guardedness on raw materials. Perhaps the most strikingly illustrative example of this was in a holding company office in Kigali where front-desk employees would not talk about raw material sources but were happy to speak about their strong international trade partners. Particularly striking about this occasion was that this exchange took place alongside display cases with gold and gemstone jewellery and under a series of back-lighted logos at the company's reception desk that included a Rwandan smelter, a Rwandan mining subsidiary, a Rwandan logistics company, and Elbit Dynamics, the Rwandan division of Israel's largest weapons manufacturer. It is important to state that the precise nature of this relationship is difficult to determine, but we can learn something important from the disclosure of such corporate links. At the very least, it tells us that there is no sensitivity or risk of scrutiny around conflict elsewhere, further along the supply chain. Displaying an Elbit logo – the Israeli military's largest supplier at a point that was 13 months into the genocidal war on Gaza – is not deemed an issue at all for a company whose entire corporate identity is built on ethical supply chains and conflict free minerals.

Piecing together this information from fieldwork and secondary materials guides our attention to carefully angled transparency where sites of extraction – i.e., the movement of minerals *in* or *to* Rwanda – are obscured yet meticulous systems are in place to enable and trace the movement of minerals *from* Rwanda. One further example sharpens the contours of this idea. Luna Smelter in Karuruma (~8 km northwest of Kigali) is held up as a model operator by the Rwandan authorities for its management's commitment to ethical mining and supply chain tracking (see e.g., The New Times, 2024). The plant has been certified by the ITSCI traceability programme and recently (in 2022) developed its own chain of custody system using Minespider's OreSource tool, a blockchain-based due diligence mechanism that securely archives the downstream provenance of mineral shipments. This initiative ensured that Luna remained compliant to the RMAP (Responsible Minerals Assurance Process), the primary assessment framework of the RMI, and grounds the claim that Luna is 'the only active, conflict-free, fully OECD-compliant and RMI recognised tin smelter in the region'.[9] This bubble is burst, however, in the simple observation that all such traceability schemes only flow one way, as the Minespider case study of Luna explicates: 'to date, there are no tools available that provide an immutable record documenting a shipment of minerals from a mine along a mineral supply chain to a smelter or refinery'.[10] The pattern of traceability flowing only downstream is in this way written into the logics of conflict free certification as a smelting site becomes something of a "point zero", a site of origin and processing removed from conflict, and a product that is, therefore, exportable as "conflict free".[11] As a consequence, in the downstream from this particular facility is a catalogue of blue chip tech companies who purchase tin from the site as part of their portfolios of responsible supply chains: Apple (2023), Ford (2020), Nokia (2024), Samsung (2022), Sony (2024), Tesla (2022).

What is obscured in this system of certification that produces transparency downstream but neglects the upstream origins of minerals? There is a crucial detail of "conflict minerals" in Rwanda that cannot be overstated: it is widely documented that large amounts of the ore that moves through the country's metallurgical infrastructures originates in the DRC. All indications are that it crosses borders to the north and south of Lake Kivu (by land and by air) from war-affected areas to be ethically "cleaned" in the Rwandan section of the supply chain. In the early 2000s, the United Nations estimated that the Rwandan army controlled 60–70 % of all coltan (tantalum ore) produced in the DRC (see Nest, 2011, pp. 88–89) and Rwandan influence, it is widely accepted, has persisted since. Though the Rwandan government claims not to be involved in the current phase of war in eastern DRC, the UN Group of Experts on the DRC (2023, 9) has documented the presence of the Rwandan Army (RDF) fighting alongside the Rwandan-backed *Mouvement du 23 mars* (M23) military group (see also BBC, 2025). The ongoing violence facilitates, the UN further documents, the smuggling of minerals across the Goma-Gisenyi border to the north of Lake Kivu. Minerals taking this route are 'laundered into the official supply chain using the International Tin Association Tin Supply Chain Initiative tags' (UN 2023, 15). In its 2021 report, the UN Group of Experts recorded that 'cross-border smuggling of untagged tantalum and tin [ores] into Rwanda continued' (UN Group of Experts on the DRC, 2021, 19) using false compartments in trucks and that Goma-based coltan and cassiterite dealers routinely transported their goods this way. Tellingly, there seems to be – purposively or not – a pronouncedly lax approach taken on the Rwandan side of the border where authorities, despite a continued flow of untagged minerals, report no detected cases of smuggling for the time period covered by this particular report (Nov 2020–April 2021) (UN Group of Experts on the DRC, 2021, 19). To add here is that the United Nations is but one source of many that document the movement of untagged minerals from DRC conflict areas into Rwanda (see e.g., Amsterdam & Partners LLP, 2024; Global Witness, 2022; Smith, 2021, 80; SIPRI, 2011, p. 9).

Once minerals have entered Rwanda, they are laundered via different practices such as "mismatched records" and "dummy mines". Mismatched records occur when exported quantities of minerals outweigh production from Rwandan mines and import records (see e.g. Eichstaedt, 2011, pp. 192–193) and so-called "dummy mines" refers to the prevalence of mining infrastructure that produces little or no ore yet is maintained to legitimise inflated claims of deposits and production (see Amsterdam & Partners LLP, 2024; Rever, 2023). At these dummy mines there are also reports that untagged ore is dumped so that it can be "re-mined" and then funnelled through official channels (see Mallinder, 2024). Taken together, the result is that the productivity of Rwandan mines is routinely inflated to account for untagged ores that arrive from war-affected areas of the DRC. The levels of this inflation are quite significant: a number of different investigations using varied sources estimate that a full 90 % of the "conflict free" minerals exported from Rwanda *as Rwandan* moves without documentation over the border with the DRC (Global Witness, 2022; Hervé, 2023). What is more, there is every reason to conclude that traceability schemes are complicit, motivated by the levy they are able to charge on shipments to allow militia-controlled mines and trade routes to arrive at "conflict free" tagging facilities over borders (Vogel, 2022, p. 113). In this way, tracing and tagging is considered a "game" (see Smith, 2021, 267–297) that appeases CSR and SEC requirements with little or no benefit for Congolese mining communities (see Vogel, 2022), and the ITSCI specifically has been labelled a "laundromat" by investigative journalists (Global Witness, 2022). Congolese politicians have claimed that $1bn per year is lost in mineral revenues to the movement of untagged minerals through Rwanda (Wilson & Schipani, 2023), a figure that fuels the accusation that "Kigali is not built with the wealth of Rwanda … [but] the resources of the DRC" (see Hervé, 2023). The DRC government has called for sanctions against Rwanda because of its involvement in war – both directly in the presence of the RDF and more indirectly in training and financial support for non-state military groups – and for the sequestering of mining sites by proxy and channelling commodities through Rwanda processing and exporting infrastructures (Wilson and Schipani, 2023).

"Conflict free" minerals are produced precisely by obscuring these links to ongoing conflict in the DRC. This is the substance of a system of

[9] https://lunasmelter.com/our-values/.

[10] https://www.minespider.com/case/luna-smelter.

[11] It is important to explicate that we do not claim that "conflict minerals" are laundered specifically through the Luna Smelter, our point is rather that such processing sites are positioned as origins of minerals in terms of traceability, whether or not mined materials are from war-affected areas.

uneven – or carefully angled – transparency where questions of provenance, or the movement of minerals *to* and *in* Rwanda, are foreclosed while meticulous records are kept that document the movement of minerals *from* Rwanda. This enables, as we have outlined here, Rwandan "conflict free" minerals to enter global networks of manufacturing in the consumer goods sold by tech companies such as Apple, Ford, Nokia, Samsung, Sony, Tesla, and so forth. In the next section we look further into supply chains downstream of Rwanda to show how "conflict free" minerals are linked back to conflict—as minerals that go to war in components of high-tech military equipment.

3. "Conflict free" minerals in weapons production

Once certified, "conflict free" minerals move into global networks of manufacturing. In this section we outline military uses of tantalum and tin from central Africa and provide specific examples of "conflict free" minerals moving from Rwanda to the arsenals of powerful militaries (e.g., those of US, UK, Israel) and therefore also to the sites they target (Afghanistan, Gaza, Iraq, etc).

It is first crucial to emphasise that militaries and governments fully recognise the necessity of technology critical minerals to the production of "defence" wares (or weapons and weapons systems). The European Commission (2016) identifies 39 different raw materials and 47 alloys that are crucial for military deployment, and the United States Department of Defense (2021, 158, 159) lists 58 'non-fuel commodity' imports it relies on 'to maintain technical dominance over adversaries'. These EU and US policy positions follow concerns raised by defence policy analysts and lobbyists that highlight a vulnerability to Chinese control of certain minerals (mainly rare earths) and a repeated line of ensuring mineral supplies do not 'unintentionally' fund black markets and human rights abuses (e.g., Aerospace Industries Association, 2023; Parthemore, 2011; Wischer, 2024). The US Government Accountability Office (2024, emphasis added) now takes the position that 'critical materials, such as titanium, tantalum, and tungsten, are *key building blocks* in many U.S. military weapon systems' and recommends stockpiling to ensure supply chain access for these minerals. The United Kingdom Government (2023), similarly, recognises its 'ability to deploy cutting-edge military capability – whether land, air, sea, space or cyberspace – is dependent on strategic materials … [that] are found in military systems ranging from the simplest firearm to F35 fighter jets and nuclear submarines'. The F-35, to focus on just one piece of equipment, has been labelled a "flying periodic table": 'the nuts and bolts that hold the plane together are made from beryllium … gallium helps amplify the radar's signal … tantalum sits in the capacitors needed for laser targeting, controls, and cockpit displays' (Abraham, 2015, p. 168). Additionally, the world's second-largest weapons manufacturer, Northrop Grumman (2023), lists an inventory of 18 "major product categories" that contain 3 TGs, including 'air and missile defense systems, Defense Electronic Systems, Precision weapons … Electronic warfare systems, Launch vehicles, Autonomous systems, Manned aircraft, Medium and large-caliber ammunition'.

What this brings to bear on the issue of "conflict minerals" is an entirely different association of military violence with the "resources" that are extracted from the earth: the produce of central and eastern Africa is not only crucial for consumer goods but also for the capacity to wage war. Returning to Rwanda, the checks and verifications put in place by EU and US legislation have enabled an array of companies to source tin and tantalum from Rwanda, as we detail in the previous section. But – as we have built towards throughout – there is more to the supply chain than consumer goods producers: a significant cohort of weapons manufacturers also source "conflict-free" minerals from Rwandan smelters. The following companies list Rwandan smelters as providers of tin: Lockheed Martin (2017) (2023 revenues: $60.8bn), the world's largest weapons producer by revenue (SIPRI, 2024); Huntington Ingalls Industries (2019) ($9.2bn, rank 17th), the largest military shipbuilder in the US; and Elbit Systems (2023) ($5.3bn, 27th), the main provider of equipment to the Israeli military. There is, in addition, a long list of sub-contractors that source tin from Rwanda: Moog Aerospace (2020), a New York-based manufacturer that holds the contract to provide the Primary Flight Control Actuation System of the Lockheed Martin-produced F-35 fighter jet (Moog, 2019); Howmet Aerospace (2022), a Pennsylvania-based components producer provides parts that 'help the F-35 Joint Strike Fighter soar' (Howmet Aerospace, 2025); and Nortech Systems (2020), a Minnesotan company whose specialist cabling systems are used in a wide range military equipment, for instance, Patriot Missiles, Paveway, F-35s, F-22s, and Tomahawk Missiles. Tantalum from Rwanda is sourced by a similar coterie of military suppliers. AeroVironment (2023) is the US military's main supplier of small drones; Frequency Electronics (2023) provides the US Department of Defense with systems that 'better link the armed forces' sensors and shooters on the battlefield'[12]; TE Connectivity (2023) supplies connectors and wires for military drone circuitry to companies such as Lockheed Martin, Thales, and BAE Systems[13]; and TTM Technologies (2023) is the largest supplier of printed circuit boards to the US military – and counts among its largest clients Lockheed Martin, Northrop Grumman, and Raytheon/RTX.

It is important to emphasise that this is only a relatively cursory survey of those companies that have submitted Specialized Disclosure Reports to the Securities and Exchange Commission according to Section 1502 of the Dodd-Frank Act. There are many avenues from "conflict free" production to weapons production that are undocumented. For instance, as far as we can determine, some major weapons producers (e.g., Boeing, Raytheon/RTX, Northrop Grumman; General Dynamics) have not submitted full inventories of suppliers, and others do not come under US jurisdiction, such as Chinese and Russian companies that would require further detailed analysis.[14] EU Conflict Minerals Regulation came into force in 2017 and required due diligence activities "from 2021" but does not require mandatory reports by companies. Consequently, major European arms companies – e.g., BAE Systems (UK), Leonardo (Italy), Thales (France), and their large networks of Europe-based suppliers – are not yet part of this story. The evidence we present, therefore, is partial yet likely illustrative of a much wider pattern of procurement of "conflict free" minerals for weapons supply chains that extends beyond those operating under the remit of the Dodd-Frank Act requirements.

What we can be sure of from currently available information, however, is that specific products – tin and tantalum – that have attained "conflict free" certification in Rwanda are subsequently put to use in contemporary warfare. Tin is a soft and malleable metal that, aside from an early application in alloyed form (bronze), is largely unsuitable for many more "traditional" metallurgical needs in warfare, such as for the dense material needs of armour and ammunition. The advanced technologies of contemporary warfare, however, are now largely dependent on tin for its ductility, electrical conductivity, and resistance to corrosion. Tin-lead solder is the most common bonding material used in weapon and aerospace products (Chen et al., 2024), and its capacity to maintain conducting properties at high temperatures makes is essential to semiconductor connection, high-end capacitors, and in fact 'electronic components in *nearly all military hardware*' (US Department of

[12] https://freqelec.com/.

[13] https://www.cleverly4braintree.com/news/james-cleverly-visits-te-connectivitys-new-site-braintree.

[14] Chinese weapons producers – including some of the largest in the world: AVIC (2023 revenues: $20.8bn, rank 8th); NORINCO ($20.5bn, 9th); CETC ($16.1bn, 10th) – are subject to CCCMC (The China Chamber of Commerce of Metals, Minerals and Chemicals Importers & Exporters) guidelines on procurement; Russian manufacturers – e.g., United Aircraft Corporation, Tactical Missiles Corporation, and Almaz-Antey Corporation (that do not disclose profits (see SIPRI, 2024)) – are not, so far as we are aware, subject to any similar regulation.

Defense, 2024; emphasis added). Aside from use in electronics, the airframe of the F-16 fighter jet – the world's most widely used fixed-wing fighter (produced by Lockheed Martin) – is fashioned with a tin-aluminium alloy (Jackson et al., 2008) and its windshields are kept frost free (essential at higher altitudes) by a system that uses a film of indium tin oxide (ITO). ITO is also used as a coating over the canopy of the F-16 (this is also the case for other fighter jets, e.g.: F-22, F-35, Mig-29) for its property of reflecting radar waves and thus enhancing the stealth capacities of the aircraft. Tin nanoparticles are also used to increase the efficiency of lithium-ion batteries[15] that are essential to the function of drones and missile guidance systems such as the widely used JDAM (Joint Direct Attack Munition) that converts standard bombs into weapons of "precision".

Tantalum is a hard, dense metal that is also crucial for in semiconductors in high performance military electronics and is therefore similarly present in nearly all military hardware. The basic circuitry of drones, notably, is built with tantalum, and their prominence in 21st Century warfare has prompted claims that the weapons industry now uses more coltan (tantalum ore) than smartphone and tablet manufacturers (see Stop the War Coalition, 2013). Tantalum is also used in counter-drone equipment developed by Lockheed (2012) for the US Army, the Enhanced Area Protection and Survivability system (EAPS). EAPS features a tantalum-tungsten alloy warhead whose density and agility enable the shooting down of, apparently, increasingly agile and evasive drone systems that are "hostile" to American targets (US Army, 2015). Tantalum is additionally a key ingredient of infrared camera tubes in night vision equipment, for example in the litening pods carried by aircraft such as the American B-52 and Israeli Hermes 450 drone, 'an electro-optical infrared sensor system [that enables] … precision targeting, close air support, intelligence, surveillance and reconnaissance'. The significance of this military capability – owed to the adsorbent qualities of tantalum – should not be understated: both the US Marine Corps and the Israeli military are on record stating the crucial advantages they maintain during operations that is owed to night vision equipment (see Johnson, 2011, p. 117). In a survey of military uses of technology critical minerals more generally, the investigative journalist, Judi Rever (2024), has made the claim that 'it is no longer possible for the US to wage battles abroad without tantalum'.

We do not claim that all of the tin and tantalum put to such military uses is procured from "conflict free" sources; there is a significant trade of both metals from outside the CAHRA region. Nor do we argue that a majority of it is sourced through the channels we document in Rwanda. But it is the case that minerals labelled "conflict free" in the CAHRA region do make their way to weapons manufacturers and to the arsenals of belligerent states – and eventually to the sites they target (Afghanistan, Gaza, Lebanon, Iran, Iraq, Ukraine, and so forth). Long-accepted ideas of "conflict minerals" and "conflict free" minerals therefore emerge in entirely different form. Conflict is newly collocated with minerals at later stages in supply chains as they once again become "conflict minerals" as part of military equipment that goes to war.

4. Conclusions

To conclude, we wish to briefly emphasise or clarify four important points before sketching out potential trajectories for future geographical research on "conflict free" minerals that go to war. First, the association of *conflict* and *minerals* examined here provokes a new vision of an important but currently narrow concept. Even the most insightful critique can – productively and meaningfully – be turned on its head, so that framings that depict the DRC as 'ground zero for conflict minerals' (see Kinniburgh, 2014, 61) or that the 'final truth' of mining in the DRC is that miners' deaths, tragically, 'count for nothing' (Kara, 2023, p. 240) now read quite differently. The DRC is in fact one of many future ground zeros at the other ends of supply chains, ones where there will be many other deaths. At different times and to different extents, Gaza, Kabul, Mosul, Fallujah, Tyre, Tripoli, Kandahar, and a great many other sites have been reduced to scenes of rubble and indiscriminate death using equipment whose destructive capacities are enhanced by "conflict free" minerals, often sourced from the DRC and surrounding areas (such as Rwanda). Second and connectedly, substantiating these connections provides a fuller account of contemporary warfare – or "late modern war" as it has been termed in political geography and international relations (e.g., Belcher, 2011; Gregory, 2011; Griffiths, 2022; Griffiths & Redwood, 2024; Jabri, 2016) – so as to include articulations between different modes, strategies, and arsenals of warfare. Too often scholars (mainly in the West) discuss "new" forms of war in a Western-centric and technology-centric manner (see Griffiths & Redwood, 2024) that forecloses wars that are fought without significant digital or aerial capabilities but yet which are, as we show, connected. Notions such as "late modern war" can thus be modified – even *decolonised* – so as to include otherwise largely overlooked (or certainly less often examined) geographies of late modern war; we begin here in the DRC eastern borderlands but other questions arise: what supply chains (e.g., through the UAE and China) are prevalent in the provision of equipment for the Sudanese Armed Forces? How are minerals moved from the DRC and into processing facilities in China and subsequently in the large but under-studied Chinese weapons industry? How do strategies of contemporary or late modern war affect mineral commodity prices and thus the stakes of conflict? Addressing these and many other questions would further understandings of contemporary warfare in ways that dislodge Western militaries as a main point of reference.

Third, lest we be misunderstood: nothing written here is geared towards advocating for better traceability schemes. Traceability should be understood as a problem of capitalism with capitalist solutions that distribute and maintain uneven conditions of exploitation and economic growth. There is no justice in a system that tends towards environmental destruction, extremely precarious labour, and shareholder capital accumulation. It is a stark indictment of the conflict free movement when its founding organisations – such as the Enough Project, whose stated mission is 'to counter genocide and crimes against humanity'[16] – call on 'the aerospace industry' to play a bigger role: 'Boeing or Lockheed Martin should signal to their suppliers that they should only be sourcing from conflict-free smelters'.[17] Clearly, nothing can be conflict free if it is led by a politics that colludes with weapons-producing companies, even as some operate (e.g., especially in the case of Boeing) at a nexus of civilian-military "dual use" technologies that troubles hard and fast definitions of what precisely constitutes a "war component" (see e.g., Jones et al., 2024). Fourth, and to reiterate, the issue of conflict and its collocation with minerals must be considered across a supply chain. As long as it is parsed off to sites and conditions of extraction, end-point users are excused of critique, with devastating consequences. To take an illustrative example, Elbit Systems (2023), a company at the centre of Israel's genocidal and ecocidal war machine (see El-Shewy, Griffiths, & Jones, 2025), is able to take a position in its supply chain policy against 'human rights abuses and atrocities' through a commitment 'to sourcing materials from companies that share our values with respect to human rights, ethics and environmental responsibility'. Need it be said, if Elbit's "values" with regards to military violence and atrocity are transposed across supply chains then all is lost. The idea of conflict (free) minerals must not be acceded to weapons manufacturers such as Elbit, Boeing, or Lockheed Martin, nor to the campaigns (e.g., Enough Project) and schemes (e.g., the RMI's RMAP and ITSCI) that legitimise their bloody business.

If we are, then, to fully reckon with the fact that "conflict free"

[15] https://www.internationaltin.org/tin-in-lithium-ion-batteries/.

[16] https://enoughproject.org/about.

[17] https://enoughproject.org/blog/29-smelters-go-conflict-free-more-help-aerospace-companies-needed.

minerals go to war, then what of future trajectories of research? An exhaustive agenda is of course beyond our capabilities but we wish to close by sketching out a set of urgent concerns for critical geographies of minerals and conflict. One example is the fact that numerous other minerals, from more "traditional" ones such as copper (which remains the base metal of all electrical circuitry) to cobalt and rare earth metals, are subject to similar conditions of extraction in Central Africa and other militarised contexts yet by virtue of a somewhat arbitrary definition of 3 TGs are not subject to similar scrutiny. Cobalt, to take one example, is crucial to the eastern DRC war economy (see Kara, 2023) and to the construction of key weapons of war (e.g., the F-16 fighter jet) (see Griffiths & Rubaii, 2025, p. 46). That legislators in Brussels and Washington have set lines around certain commodities ought to be prompt enough to look elsewhere to develop more comprehensive understandings of the breadth of minerals that are newly essential to contemporary warfare. Another area of focus should undoubtedly be the roles of China and Russia in the mining, processing, and transportation of technology critical minerals, as well as the growing prominence of Chinese and Russian weapons manufacturers of which we currently know very little (at least in anglophone critical scholarship). This is no small task: in a research area characterised by security and secrecy, producing knowledge on the role of a notoriously closed state-industrial complex presents great challenges but, having examined many supply chain disclosure reports, the role of Chinese mineral suppliers especially is too significant to be overlooked. Geopolitical issues of stockpiling, tariffs, and embargoes, of course, are likely another part of the picture, especially in an era of more fractious relations between the US and China that are likely to influence ongoing wars in the DRC borderlands and the destinations of minerals that are central to those conflicts and, as we have shown, maintaining "defense superiority" (see Vivoda et al., 2025). A final issue, a most important one, is to trace the environmental and public health effects that are prevalent and shared between sites of extraction and sites of bombardment – and thus also likely at logistical and manufacturing sites. Minerals such as tin, tungsten, and tantalum – and also cobalt, copper, lithium, and so forth – are extractable only with great chemotoxic burden (Atibu et al., 2013; Banza Lubaba Nkulu et al., 2018; Van Brusselen et al., 2020) and with ecological and health effects that are repeated when those minerals are deployed in warfare (see e.g., Abuawad et al., 2025; Griffiths & Rubaii, 2025). Both in terms of uneven toxic exposures and labour exploitation, there is a pervasive and pernicious racial politics to conflict (free) minerals that is yet to be fully explored and countered.

Counter is an appropriate theme to close on. In this approach to conflict and minerals is an explication of supply chains that make war possible. A major potential of research into "conflict free" minerals that go to war is to explicate lines of international responsibility and complicity in warfare that does not merely refresh research agendas but also identifies strategic sites of accountability and intervention. As is clear in the research presented here, there is a global network of mining, logistics, and smelting actors to be considered within the already vast military industrial complex that supports war. In presenting this research, our practical aim, at its most ambitious, is to establish routes to severing that support.

CRediT authorship contribution statement

Kali Rubaii: Writing – original draft, Conceptualization. **Mohamed El-Shewy:** Writing – original draft, Conceptualization. **Mark Griffiths:** Writing – original draft, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest.

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Data will be made available on request.

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